

TrES-3b

R-1976

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_131004 · created 2026-08-07T00:21:21+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456569.70297 +/- 0.00082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.015
Transit depth (Rp/Rs)^2	3.86 +/- 0.57 [%]
Orbital Inclination (inc)	82.77 +/- 0.76
Airmass coefficient 1 (a1)	1.016 +/- 0.019
Airmass coefficient 2 (a2)	-0.006 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.9 %
Best Comparison Star	#2 - [457, 143]
Optimal Aperture	2.94
Optimal Annulus	6.84
Transit Duration (day)	0.065 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.015 vs 0.16309 (deviation 2.19σ)
Duration fit vs published	0.065 d vs 0.0567 d (deviation 1.54σ)
Mid-time O–C	2.0 min
Depth SNR	13.1
Residual scatter	1.9 %

Light curve

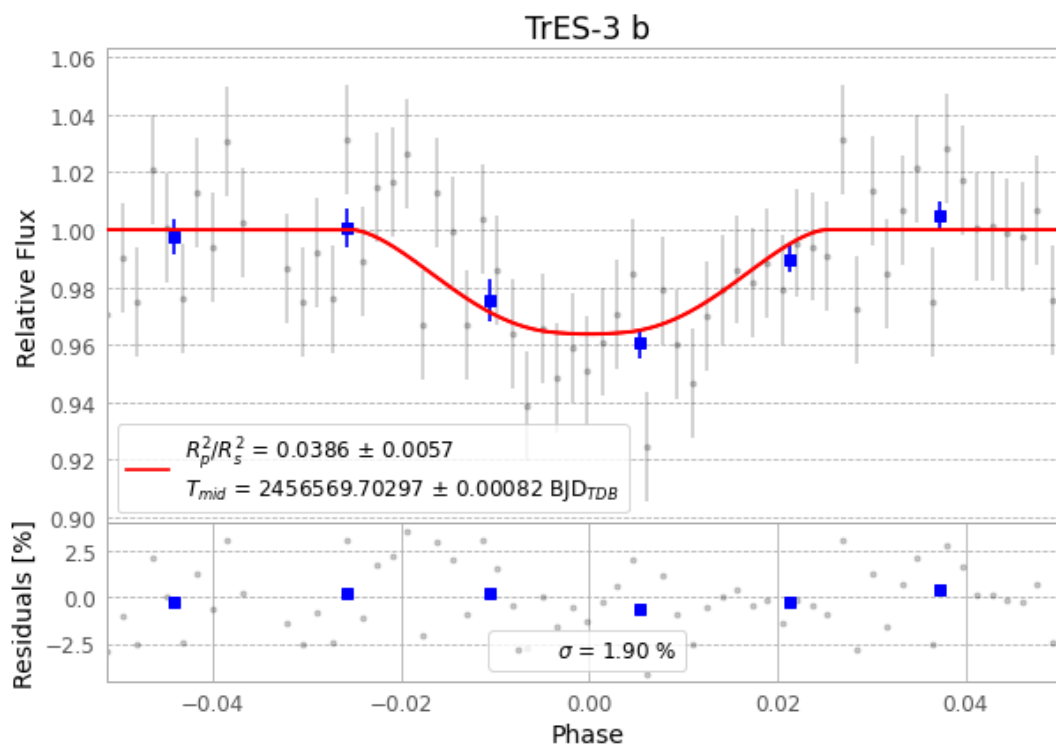


Figure 1 — FinalLightCurve_TrES-3 b_2013-10-03.png (SHA-256 30f2a2e2163f4184...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [252, 254], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4eb0cbdb3658142b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

65 FITS frames (43 MB), TRES-3131004031513.FITS → TRES-3131004062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-131004.log (7531afa98ef1...), FinalLightCurve_TrES-3 b_2013-10-03.png (30f2a2e2163f...), FinalLightCurve_TrES-3 b_2013-10-03.csv (48e91d3a24b5...)

Compute resources

Wall time 97.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:21Z.